

## MULTIVARIABLE POTENTIAL THEORY • DENNIS G. ZILL &amp; DR. HAGHIGHAT LECTURE 4

## Chapter 2.4: Exact Differential Equations

Differential forms, Clairaut's mixed partials test, potential functions  $f(x, y) = C$ , and the systematic 5-step reconstruction protocol.

Zill §2.4 (pp. 89–98)

Lecture 4 Slides 3–9

Midterm 1 Must-Know

**PLAIN-ENGLISH INTUITION**

An equation  $M(x, y)dx + N(x, y)dy = 0$  is **exact** if it is the total differential  $df = 0$  of a hidden surface  $f(x, y) = C$ . Because mixed second partials of smooth functions are always identical ( $\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$ ), we test exactness by checking if  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ . If they match, partial integration reconstructs the exact implicit solution!

**THEOREM 2.4.1 • CRITERION FOR AN EXACT DIFFERENTIAL**

Let  $M, N, M_y, N_x$  be continuous in an open rectangle  $R$ . The differential form  $M(x, y)dx + N(x, y)dy$  is **exact** if and only if:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Then there exists  $f(x, y)$  with  $\frac{\partial f}{\partial x} = M$  and  $\frac{\partial f}{\partial y} = N$ . General solution:  $f(x, y) = C$ .

**The 5-Step Reconstruction Protocol**

1. **Test:** Verify  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ .
2. **Integrate  $M$ :**  $f(x, y) = \int M(x, y) dx + g(y)$ .
3. **Differentiate w.r.t.  $y$ :**  $\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \int M dx + g'(y)$ .
4. **Equate to  $N$ :** Solve for  $g'(y)$  (all  $x$ 's must cancel!). Integrate  $g'(y)$  to find  $g(y)$ .
5. **Assemble:** State implicit curve  $f(x, y) = C$ .

**Master Worked Example 1 • Full 5-Step Reconstruction (Lecture 4 Slide 8)**

CONCORDIA SLIDE EX

**Problem:** Solve the initial value problem:  $(2xy)dx + (x^2 - 1)dy = 0$ , with  $y(0) = 3$ .

**1 Test Exactness:**

$$M = 2xy \implies \frac{\partial M}{\partial y} = 2x.$$

$$N = x^2 - 1 \implies \frac{\partial N}{\partial x} = 2x.$$

$$M_y = N_x \implies \text{EXACT!}$$

**2 Integrate  $M$  w.r.t  $x$ :**

$$f(x, y) = \int 2xy dx + g(y) = x^2 y + g(y).$$

**3 Differentiate w.r.t  $y$  & Equate to  $N$ :**

$$\frac{\partial f}{\partial y} = x^2 + g'(y) = x^2 - 1 \implies g'(y) = -1.$$

$$\text{Integrate: } g(y) = \int (-1) dy = -y.$$

**4 Assemble  $f(x, y) = C$  & Apply  $y(0) = 3$ :**

$$x^2 y - y = C \implies y(x^2 - 1) = C.$$

$$y(0) = 3 \implies 3(-1) = C \implies C = -3.$$

$$\text{Explicit Solution: } y(x) = \frac{3}{1-x^2}.$$

## 2 Rescuing Non-Exact Equations with Integrating Factors

### RESCUE FORMULAS

When  $M_y \neq N_x$ , we multiply the entire equation by  $\mu$  to make it exact. We check whether the discrepancy  $\frac{M_y - N_x}{N}$  is a function of  $x$  alone, or if  $\frac{N_x - M_y}{M}$  is a function of  $y$  alone.

#### CASE 1: FACTOR $M(X)$ (FUNCTION OF $X$ ALONE)

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N(x, y)} = p(x) \implies \mu(x) = e^{\int p(x) dx}$$

#### CASE 2: FACTOR $M(Y)$ (FUNCTION OF $Y$ ALONE)

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M(x, y)} = q(y) \implies \mu(y) = e^{\int q(y) dy}$$

### Master Worked Example 2 · Non-Exact Rescue (Lecture 4 Slide 10–11)

CONCORDIA SLIDE EX

**Problem:** Solve  $xy dx + (2x^2 + 3y^2 - 20) dy = 0$ .

#### 1 Test Exactness:

$$M = xy \implies M_y = x.$$

$$N = 2x^2 + 3y^2 - 20 \implies N_x = 4x.$$

$$M_y \neq N_x \implies \text{NOT EXACT. Discrepancy: } M_y - N_x = -3x.$$

#### 2 Find Integrating Factor:

$$\bullet \frac{M_y - N_x}{N} = \frac{-3x}{2x^2 + 3y^2 - 20} \text{ (not pure } x\text{).}$$

$$\bullet \frac{N_x - M_y}{M} = \frac{4x - x}{xy} = \frac{3x}{xy} = \frac{3}{y} \text{ (pure } y\text{!).}$$

$$\mu(y) = e^{\int \frac{3}{y} dy} = e^{3 \ln y} = y^3$$

#### 3 Multiply DE by $\mu(y) = y^3$ :

$$(xy^4) dx + (2x^2y^3 + 3y^5 - 20y^3) dy = 0.$$

$$\text{Verify: } M_y = 4xy^3, \quad N_x = 4xy^3 \text{ (EXACT!).}$$

#### 4 Reconstruct $f(x, y)$ :

$$f(x, y) = \int xy^4 dx + g(y) = \frac{1}{2}x^2y^4 + g(y).$$

$$\frac{\partial f}{\partial y} = 2x^2y^3 + g'(y) = 2x^2y^3 + 3y^5 - 20y^3.$$

$$g'(y) = 3y^5 - 20y^3 \implies g(y) = \frac{1}{2}y^6 - 5y^4.$$

$$\text{Final Solution: } \frac{1}{2}x^2y^4 + \frac{1}{2}y^6 - 5y^4 = C.$$

### THE CANCELLATION LITMUS TEST & SIGN MEMORY

- In Step 4, if any  $x$  terms fail to cancel in  $g'(y)$ , an error occurred earlier!
- Remember the sign inversion: When dividing by  $M$ , the numerator is  $N_x - M_y$  (reversed!).